

KHIZAR ANJUM
Machine Learning Engineer

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137 citations, h-index: 7, i10-index: 7



PROFESSIONAL EXPERIENCE

Graduate Research Assistant, CPS Lab, Rutgers University — (2019 – Present) | Piscataway, NJ, US

- Specialized in developing efficient deep learning architectures for various applications, with focus on **resource-constrained devices**. Published **4 journal articles** and **14 conference papers** in prestigious venues.
- Filed **3 patents** and helped secure funding worth **\$1.135 million** in prestigious NSF awards.

Research Assistant, Communications Lab, LUMS — (2017 – 2019) | Lahore, Pakistan

- Developed and trained cutting-edge **Convolutional Neural Network classifier** with **Bi-LSTM evidence aggregation** model for Parkinson's detection on **multimodal data**.
- Achieved **89% Area under ROC Curve performance** compared to baseline of 56%, a **23% improvement**.

RESEARCH INTERESTS

Analog and Neuromorphic Computing, Ultra-Low Power Neural Networks, Hardware-Software Co-design, FPGA Acceleration, Embedded Machine Learning, Computer Vision, Signal Processing, Underwater Acoustic Communications, Autonomous UAVs

LEADERSHIP EXPERIENCE

- Served as Program Coordinator for **two years** on ECE Department's **first-ever Graduate Student Association (GSA)**.
- Mentored **two** Capstone groups who got **first place** and **best-for-commercialization** awards at Rutgers University.

EDUCATION

Rutgers University, New Brunswick, NJ, (2019–2026 [Expected])

- PhD Candidate & Master of Science (MS), Electrical and Computer Engineering (ECE)
- Grade Point Average (GPA): 3.92/4; *Advised by* Dr. Dario Pompili

Lahore University of Management Sciences (LUMS), Lahore, Pakistan (2015–2019)

- Bachelor of Science (BS), Electrical Engineering (EE) – Gold Medalist
- Grade Point Average (GPA): 3.86/4; *Advised by* Dr. Muhammad Tahir and Dr. Momin Uppal

SELECTED RESEARCH PROJECTS

Ultra-Low Power Analog Neural Network Design for Health Monitoring — 2022-25

- Scaled existing VRPU design by writing a **custom Python-based generation script** for ngspice circuit simulator.
- **Invented a novel analog architecture** called CvPU (Convolutional Processing Unit) that uses anisotropic diffusion to implement a convolutional layer with a **8-DoF 3×3 kernel** when tessellated, consuming only **55.58 mW**.
- Tested and implemented the digital-side neural networks on FPGAs such as TinyFPGA BX and Xilinx Ultra96-v2 by **writing custom neural computation modules** using Chisel hardware construction language.

Crowd Prediction and Behavior Assessment Using Adaptive UAVs — 2024-25

- **Leveraged efficient signal processing techniques** by piggy-backing inter-prediction results from MPEG-4 hybrid video-encoder for crowd pattern identification to deliver **45× reduction** in execution time compared to baseline.
- Implemented **automated exponentially converging implementation** of Volkov Block Method for solving Laplace Equation, identifying crowd behavior based on domain. <https://github.com/khizar-anjum/pyBlockMethod>

Deep Joint Source Channel Coding for Underwater Image & Video Transmission — 2022-25

- Implemented a variable encoder that **outperforms traditional JPEG/JPEG2000 approaches** for compression in low-SNR regimes, **i.e., from 0 to 13 dB, and avoids cliff-effects** observed in brittle traditional techniques.
- Collected **first-ever real-life acoustic communication dataset** under various center frequency, bandwidth, and TX/RX configurations at Barnaget Bay, NJ and Atlantic City, NJ and **published the dataset as ACommSet**.

RELEVANT SKILLS

- Python (advanced), C++, Chisel HDL, MATLAB, Bash, SPICE (ngspice), PyTorch, TensorFlow, Megatron-LM, Scikit-learn, OpenCV, Pandas, Xilinx Vivado, PYNQ, TinyFPGA, Chisel language, FPGA deployment, GNU Radio, ROS, AirSim, low-power analog-digital ML systems, SLURM, NERSC Perlmutter, distributed training with parallelism techniques, Docker, Git/GitHub, Linux, VS Code, Jupyter, Model Context Protocol (MCP)

ACHIEVEMENTS & AWARDS

- Recipient of **Rutgers Innovation Awards** in the Graduate Student Category, 2025
- Recipient of **Paul Panayotatos Scholarship** for Sustainability-aware Electrical Engineering, 2024
- Selected for **NSF NOVUS & NSF I-Corps** program for innovation & customer discovery training, 2022
- Recognized with **Gold Medal** at LUMS for academic excellence, 2019
- Won **TA Achievement Award**, 2020 from Rutgers